Analytical Derivations for Digital Quantum Simulation of a Two-Qubit Transverse-Field Ising Model

Yulan Fan

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Abstract

This note collects the hand calculations underlying a three-notebook implementation of exact and Trotterized real-time dynamics for a two-qubit transverse-field Ising model. The derivations proceed from basis states and tensor-product operators to the exact spectrum and time evolution, and then to first- and second-order product formulas implemented with RX and RZZ gates. The emphasis is on transparent operator algebra, independent validation, and the distinction between state error, infidelity, observable error, and circuit cost.

Contents

1 Model and conventions

We study the two-qubit Hamiltonian

$$H = -JZ_0Z_1 - h(X_0 + X_1), \tag{1}$$

where J is the Ising coupling and h is the transverse-field strength. We use units with $\hbar = 1$ and the ordered computational basis

$$(|00\rangle, |01\rangle, |10\rangle, |11\rangle). \tag{2}$$

The one-qubit basis vectors and Pauli matrices are

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \tag{3}$$

$$X = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}, \qquad Y = \begin{pmatrix}0 & -\mathrm{i}\\\mathrm{i} & 0\end{pmatrix}, \qquad Z = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}. \tag{4}$$

The two-qubit basis vectors are

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}. \tag{5}$$

2 Two-qubit operators and Hamiltonian matrix

The local operators are

$$X_0 = X \otimes \mathbb{I}, \qquad X_1 = \mathbb{I} \otimes X, \tag{6}$$

$$Z_0 = Z \otimes \mathbb{I}, \qquad Z_1 = \mathbb{I} \otimes Z, \tag{7}$$

and

$$Z_0Z_1 = Z \otimes Z. \tag{8}$$

Their explicit matrices are

$$X_0 = \begin{pmatrix}0 & 0 & 1 & 0\\0 & 0 & 0 & 1\\1 & 0 & 0 & 0\\0 & 1 & 0 & 0\end{pmatrix}, \qquad X_1 = \begin{pmatrix}0 & 1 & 0 & 0\\1 & 0 & 0 & 0\\0 & 0 & 0 & 1\\0 & 0 & 1 & 0\end{pmatrix}, \tag{9}$$

$$Z_0Z_1 = \mathrm{diag}(1, -1, -1, 1). \tag{10}$$

A matrix can be reconstructed from its action on an ordered basis: the jth column is the vector obtained by applying the operator to the jth basis state. For example,

$$X_0|00\rangle = |10\rangle, \quad X_0|01\rangle = |11\rangle, \quad X_0|10\rangle = |00\rangle, \quad X_0|11\rangle = |01\rangle, \tag{11}$$

which immediately gives the columns of X_0.

Substituting these operators into Eq. (1) gives

$$H = \begin{pmatrix}-J & -h & -h & 0\\-h & J & 0 & -h\\-h & 0 & J & -h\\0 & -h & -h & -J\end{pmatrix}. \tag{12}$$

The matrix is Hermitian, as required for a physical Hamiltonian.

3 Symmetry-adapted basis and exact spectrum

Introduce the symmetric and antisymmetric combinations

$$|\phi_\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\psi_\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{13}$$

The antisymmetric states are eigenstates of the full Hamiltonian. Direct application gives

$$H |\phi_-\rangle = -J |\phi_-\rangle, \tag{14}$$

$$H |\psi_-\rangle = +J |\psi_-\rangle. \tag{15}$$

The remaining symmetric subspace is spanned by $\{|\phi_+\rangle, |\psi_+\rangle\}$. In this basis,

$$H_{\text{sym}} = \begin{pmatrix} -J & -2h \\ -2h & J \end{pmatrix}. \tag{16}$$

Define

$$\Omega = \sqrt{J^2 + 4h^2}. \tag{17}$$

Since

$$H_{\text{sym}}^2 = \Omega^2 \mathbb{I}, \tag{18}$$

the two eigenvalues of this block are $\pm\Omega$. The full spectrum is therefore

$$\boxed{E \in \{-\Omega, -J, J, \Omega\}}. \tag{19}$$

4 Exact time evolution from $|00\rangle$

The initial state decomposes as

$$|00\rangle = \frac{|\phi_+\rangle + |\phi_-\rangle}{\sqrt{2}}. \tag{20}$$

The antisymmetric component evolves trivially:

$$e^{-iHt} |\phi_-\rangle = e^{iJt} |\phi_-\rangle. \tag{21}$$

For the symmetric block, the identity $H_{\text{sym}}^2 = \Omega^2 \mathbb{I}$ gives

$$e^{-iH_{\text{sym}}t} = \cos(\Omega t)\mathbb{I} - i\frac{\sin(\Omega t)}{\Omega} H_{\text{sym}}. \tag{22}$$

Acting on $|\phi_+\rangle$,

$$e^{-iH_{\text{sym}}t} |\phi_+\rangle = \left[\cos(\Omega t) + i\frac{J}{\Omega}\sin(\Omega t)\right] |\phi_+\rangle + i\frac{2h}{\Omega}\sin(\Omega t) |\psi_+\rangle. \tag{23}$$

Hence

$$\begin{aligned} |\psi(t)\rangle = \frac{1}{\sqrt{2}} \Bigg\{ & e^{iJt} |\phi_-\rangle + \left[\cos(\Omega t) + i\frac{J}{\Omega}\sin(\Omega t)\right] |\phi_+\rangle \\ & + i\frac{2h}{\Omega}\sin(\Omega t) |\psi_+\rangle \Bigg\}. \end{aligned} \tag{24}$$

Returning to the computational basis gives

$$\boxed{\begin{aligned} a_{00}(t) &= \frac{1}{2}\left[\mathrm{e}^{\mathrm{i}Jt} + \cos(\Omega t) + \mathrm{i}\frac{J}{\Omega}\sin(\Omega t)\right], \\ a_{11}(t) &= \frac{1}{2}\left[-\mathrm{e}^{\mathrm{i}Jt} + \cos(\Omega t) + \mathrm{i}\frac{J}{\Omega}\sin(\Omega t)\right], \\ a_{01}(t) &= a_{10}(t) = \mathrm{i}\frac{h}{\Omega}\sin(\Omega t). \end{aligned}} \tag{25}$$

The basis probabilities are $P_{ij}(t) = |a_{ij}(t)|^2$. In particular,

$$P_{01}(t) = P_{10}(t) = \frac{h^2}{\Omega^2}\sin^2(\Omega t). \tag{26}$$

5 Observables

The longitudinal magnetization per spin is

$$M_z = \frac{Z_0 + Z_1}{2}. \tag{27}$$

It has eigenvalues $+1, 0, 0, -1$ on the ordered computational basis, so

$$\boxed{\langle M_z \rangle = P_{00} - P_{11}}. \tag{28}$$

The nearest-neighbor spin correlation is

$$C_{zz} = Z_0 Z_1, \tag{29}$$

with eigenvalues $+1, -1, -1, +1$, hence

$$\boxed{\langle Z_0 Z_1 \rangle = P_{00} - P_{01} - P_{10} + P_{11}}. \tag{30}$$

These observables compress the information in the full state. For example, $\langle M_z \rangle = 0$ does not determine a unique quantum state.

6 Why the Hamiltonian terms cannot be factorized exactly

Split the Hamiltonian into

$$H = H_{ZZ} + H_X, \tag{31}$$

where

$$H_{ZZ} = -J Z_0 Z_1, \qquad H_X = -h(X_0 + X_1). \tag{32}$$

Using $[Z, X] = 2\mathrm{i}Y$,

$$[Z_0 Z_1, X_0] = 2\mathrm{i} Y_0 Z_1, \tag{33}$$

$$[Z_0 Z_1, X_1] = 2\mathrm{i} Z_0 Y_1. \tag{34}$$

Therefore

$$\boxed{[H_{ZZ}, H_X] = 2\mathrm{i}Jh\,(Y_0 Z_1 + Z_0 Y_1) \neq 0.} \tag{35}$$

Consequently,

$$\mathrm{e}^{-\mathrm{i}(H_{ZZ}+H_X)t} \neq \mathrm{e}^{-\mathrm{i}H_{ZZ}t}\mathrm{e}^{-\mathrm{i}H_X t} \tag{36}$$

for a finite time interval in general.

As a state-level check,

$$[Z_0 Z_1, X_0 + X_1]\,|10\rangle = 2\,|00\rangle + 2\,|11\rangle. \tag{37}$$

This is consistent with the operator identity above, but a single input-output pair does not uniquely determine the full operator.

7 Useful exponential identities

If an operator P satisfies $P^2 = \mathbb{I}$, then its exponential separates into even and odd powers:

$$\begin{aligned} e^{iaP} &= \mathbb{I} + iaP + \frac{(iaP)^2}{2!} + \cdots && (38)\\ &= \cos(a)\mathbb{I} + i\sin(a)P. && (39) \end{aligned}$$

This applies to every Pauli operator and to Z_0Z_1. Thus

$$e^{iaX} = \cos(a)\mathbb{I} + i\sin(a)X, \tag{40}$$

$$e^{iaZ_0Z_1} = \cos(a)\mathbb{I} + i\sin(a)Z_0Z_1. \tag{41}$$

Since Z_0Z_1 has eigenvalues $+1, -1, -1, +1$,

$$e^{iaZ_0Z_1} = \operatorname{diag}(e^{ia}, e^{-ia}, e^{-ia}, e^{ia}). \tag{42}$$

The minus sign of an eigenvalue enters the exponent; it is not an overall prefactor.

8 First-order Lie-Trotter formula

For a total evolution time t divided into r steps,

$$\Delta t = \frac{t}{r}. \tag{43}$$

The first-order approximation used here is

$$\boxed{U_1(t;r) = \left[e^{-iH_{ZZ}\Delta t}e^{-iH_X\Delta t}\right]^r.} \tag{44}$$

A Taylor expansion shows that the exact and factorized short-time evolutions agree through first order in Δt; their difference begins at $\mathcal{O}(\Delta t^2)$ and is controlled by the commutator. Repeating r steps over fixed total time gives a typical global operator/state error of order

$$\mathcal{O}\left(\frac{t^2}{r}\right). \tag{45}$$

8.1 Gate mapping

Because $[X_0, X_1] = 0$,

$$e^{-iH_X\Delta t} = e^{ih\Delta t X_0}e^{ih\Delta t X_1}. \tag{46}$$

Qiskit defines

$$RX(\theta) = e^{-i\theta X/2}, \tag{47}$$

so each transverse-field rotation uses

$$\boxed{\theta_X = -2h\Delta t.} \tag{48}$$

Similarly,

$$e^{-iH_{ZZ}\Delta t} = e^{iJ\Delta t Z_0Z_1}, \tag{49}$$

while

$$RZZ(\theta) = e^{-i\theta Z_0Z_1/2}, \tag{50}$$

which gives

$$\boxed{\theta_{ZZ} = -2J\Delta t.} \tag{51}$$

8.2 One first-order step acting on $|00\rangle$

Define

$$c = \cos(h\Delta t), \qquad s = \sin(h\Delta t), \tag{52}$$

$$p = \mathrm{e}^{\mathrm{i}J\Delta t}, \qquad q = \mathrm{e}^{-\mathrm{i}J\Delta t}. \tag{53}$$

The transverse-field part gives

$$\mathrm{e}^{-\mathrm{i}H_X\Delta t}|00\rangle = (c|0\rangle + \mathrm{i}s|1\rangle) \otimes (c|0\rangle + \mathrm{i}s|1\rangle) \tag{54}$$

$$= c^2|00\rangle + \mathrm{i}cs|01\rangle + \mathrm{i}cs|10\rangle - s^2|11\rangle. \tag{55}$$

The ZZ evolution then adds phases according to parity:

$$\boxed{U_{1,\mathrm{step}}|00\rangle = pc^2|00\rangle + \mathrm{i}qcs|01\rangle + \mathrm{i}qcs|10\rangle - ps^2|11\rangle.} \tag{56}$$

The ZZ gate does not change probabilities immediately, but its relative phases affect subsequent interference after later noncommuting rotations.

9 Second-order symmetric Suzuki-Trotter formula

The symmetric second-order step is

$$\boxed{S_2(\Delta t) = \mathrm{e}^{-\mathrm{i}H_X\Delta t/2}\mathrm{e}^{-\mathrm{i}H_{ZZ}\Delta t}\mathrm{e}^{-\mathrm{i}H_X\Delta t/2}.} \tag{57}$$

To verify its order, define $A = -\mathrm{i}H_X$ and $B = -\mathrm{i}H_{ZZ}$. Then

$$S_2(\Delta t) = \mathrm{e}^{A\Delta t/2}\mathrm{e}^{B\Delta t}\mathrm{e}^{A\Delta t/2}. \tag{58}$$

Expanding each exponential through second order and multiplying while preserving operator order gives

$$S_2(\Delta t) = \mathbb{I} + (A+B)\Delta t \tag{59}$$

$$+ \frac{1}{2}\left(A^2 + AB + BA + B^2\right)\Delta t^2 + \mathcal{O}(\Delta t^3). \tag{60}$$

Since

$$(A+B)^2 = A^2 + AB + BA + B^2, \tag{61}$$

this agrees with

$$\mathrm{e}^{(A+B)\Delta t} = \mathbb{I} + (A+B)\Delta t + \frac{1}{2}(A+B)^2\Delta t^2 + \mathcal{O}(\Delta t^3). \tag{62}$$

Therefore the local error is $\mathcal{O}(\Delta t^3)$, and for fixed total time the typical global operator/state error is

$$\mathcal{O}\left(\frac{t^3}{r^2}\right). \tag{63}$$

9.1 Gate mapping

The two half-field evolutions use

$$\boxed{\theta_{X,\mathrm{half}} = -h\Delta t,} \tag{64}$$

whereas the central interaction uses

$$\boxed{\theta_{ZZ} = -2J\Delta t.} \tag{65}$$

One symmetric step is therefore represented chronologically in a circuit by

$$RX_0(-h\Delta t)\, RX_1(-h\Delta t)\, RZZ(-2J\Delta t)\, RX_0(-h\Delta t)\, RX_1(-h\Delta t). \tag{66}$$

9.2 One second-order step acting on $|00\rangle$

Let

$$c = \cos\left(\frac{h\Delta t}{2}\right), \qquad s = \sin\left(\frac{h\Delta t}{2}\right), \tag{67}$$

and again define $p = \mathrm{e}^{\mathrm{i}J\Delta t}$ and $q = \mathrm{e}^{-\mathrm{i}J\Delta t}$. After the first half-field step,

$$|\chi_1\rangle = c^2 |00\rangle + \mathrm{i}cs\,|01\rangle + \mathrm{i}cs\,|10\rangle - s^2 |11\rangle\,. \tag{68}$$

After the full ZZ step,

$$|\chi_2\rangle = pc^2 |00\rangle + \mathrm{i}qcs\,|01\rangle + \mathrm{i}qcs\,|10\rangle - ps^2 |11\rangle\,. \tag{69}$$

Applying the second half-field step and collecting equal basis states gives

$$a_{00} = p(c^4 + s^4) - 2qc^2s^2, \tag{70}$$
$$a_{01} = a_{10} = \mathrm{i}(p+q)cs(c^2 - s^2), \tag{71}$$
$$a_{11} = -2(p+q)c^2s^2. \tag{72}$$

Use

$$p + q = 2\cos(J\Delta t), \tag{73}$$
$$2cs = \sin(h\Delta t), \qquad c^2 - s^2 = \cos(h\Delta t), \tag{74}$$

and

$$4c^2s^2 = \sin^2(h\Delta t). \tag{75}$$

The compact result is

$$\boxed{\begin{aligned} a_{00} &= \cos(J\Delta t)\cos^2(h\Delta t) + \mathrm{i}\sin(J\Delta t), \\ a_{01} = a_{10} &= \mathrm{i}\cos(J\Delta t)\sin(h\Delta t)\cos(h\Delta t), \\ a_{11} &= -\cos(J\Delta t)\sin^2(h\Delta t). \end{aligned}} \tag{76}$$

The equality $a_{01} = a_{10}$ follows from exchange symmetry and is also a useful debugging condition.

9.3 Limiting-case checks

If $h = 0$, Eq. (76) becomes

$$|\psi\rangle = \mathrm{e}^{\mathrm{i}J\Delta t}\,|00\rangle\,. \tag{77}$$

If $J = 0$, it becomes the exact product of two independent X rotations:

$$|\psi\rangle = \cos^2(h\Delta t)\,|00\rangle + \mathrm{i}\sin(h\Delta t)\cos(h\Delta t)(|01\rangle + |10\rangle) \tag{78}$$
$$- \sin^2(h\Delta t)\,|11\rangle\,. \tag{79}$$

The amplitudes also satisfy

$$|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1. \tag{80}$$

10 Error measures and convergence

For normalized pure states, the fidelity is

$$F(\psi_{\text{exact}}, \psi_{\text{approx}}) = \left|\langle \psi_{\text{exact}} | \psi_{\text{approx}} \rangle\right|^2. \tag{81}$$

The infidelity is $1 - F$. A global phase leaves the fidelity equal to one even though direct component-wise comparison may fail.

For a first-order product formula, the typical state/operator error scales as r^{-1}; for the symmetric second-order formula it scales as r^{-2}. In a generic small-error regime, the infidelity is quadratic in the perpendicular state error, leading commonly to

$$1 - F_1 \sim r^{-2}, \qquad 1 - F_2 \sim r^{-4}. \tag{82}$$

These are asymptotic expectations rather than guarantees at every coarse step size or special evolution time. Coherent cancellation can make finite-step behavior non-monotonic.

Observable errors are defined separately, for example

$$\Delta M_z = \left|\langle M_z \rangle_{\text{approx}} - \langle M_z \rangle_{\text{exact}}\right|. \tag{83}$$

A small error in one observable does not imply that the complete statevector is accurate.

11 Circuit cost

A first-order step contains two RX gates and one RZZ gate. An explicit second-order step contains four RX gates and one RZZ gate. Therefore, comparing methods at equal step count is not the same as comparing them at equal circuit cost.

In consecutive second-order steps, adjacent half-field rotations can be combined:

$$RX(-h\Delta t)RX(-h\Delta t) = RX(-2h\Delta t). \tag{84}$$

Before hardware execution, further decomposition and transpilation would be required, and the cost of RZZ depends on the available native gate set and connectivity.

12 Numerical reference point

For the values used frequently in the notebooks,

$$J = 1.0, \qquad h = 0.7, \tag{85}$$

we find

$$\Omega = \sqrt{1 + 4(0.7)^2} \approx 1.72046505, \tag{86}$$

and the exact spectrum is approximately

$$(-1.72046505, -1, 1, 1.72046505). \tag{87}$$

For one symmetric second-order step with $\Delta t = 1$,

$$|\psi_{S_2}\rangle \approx (0.31606797 + 0.84147098\mathrm{i})\,|00\rangle + 0.26622038\mathrm{i}\,|01\rangle + 0.26622038\mathrm{i}\,|10\rangle - 0.22423433\,|11\rangle. \tag{88}$$

This reference value is useful for checking parameter ordering and gate-angle conventions.

13 Implementation checklist

A reliable notebook workflow should include the following independent checks:

1. Verify the action of each tensor-product operator on all four basis states.
2. Confirm that $H = H^\dagger$.
3. Compare numerical eigenvalues with $(-\Omega, -J, J, \Omega)$.
4. Compare exact matrix evolution with Eq. (25).
5. Compare one-step Qiskit statevectors with Eqs. (56) and (76).
6. Check normalization and exchange symmetry $a_{01} = a_{10}$.
7. Print the actual RX and RZZ angles when debugging.
8. Distinguish a true component mismatch from an irrelevant global phase by computing fidelity.
9. Restart the notebook kernel and run all cells before committing results.

14 Summary

The two-qubit model is small enough to solve exactly but already contains the central ingredients of digital quantum simulation: noncommuting Hamiltonian terms, tensor-product operators, basis-dependent measurements, product-formula errors, and a trade-off between accuracy and circuit resources. The exact solution provides a controlled reference against which the gate-level approximations can be validated. The first-order formula supplies the simplest digital decomposition, while the symmetric second-order formula demonstrates how time-symmetric operator ordering cancels the leading local error.